01 / Lecture map and tutorial relevance

Gas diffusion & gas transport | Concise English edition | Lecture PDF: 19 pages; Tutorial PDF: 1 page

The lecture in one pathway

Environmental pressure → gas partial pressure → membrane diffusion → blood O_2 content → tissue O_2 delivery / unloading → CO_2 carriage → pulmonary CO_2 unloading.

Topic	Notes page	Original tutorial questions
Dalton + Henry; diving	2	Q13
Altitude + hyperventilation	3	Q4, Q7: cross-topic
Fick's law + respiratory barrier	4–5	Lecture mechanisms and in-slide exercise
O_2 content + delivery	6	Q3
Anemia, cyanosis, hypoxemia / hypoxia	7	Q6: application
Dissociation curve, P_{50} and shifts	8–9	Foundation for Q5 and Q10
Pulmonary loading + CO poisoning	10	Q5, Q10
CO_2 forms + carbonic anhydrase	11	Q12: renal mechanism is an extension
Tissue/lung exchange + Bohr/Haldane	12	Integrated CO_2 reasoning

Questions mainly belonging to other lectures

Question	Required topic
Q1	Compliance, recoil and work of breathing.
Q2	Regional V/Q ratios: apex versus base.
Q8, Q9	Airway resistance; surfactant, surface tension and lung mechanics.
Q11, Q14	Chemoreceptors and ventilatory control; dead space and alveolar ventilation.

The question pattern to watch

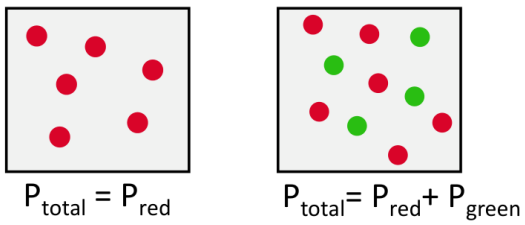
- Separate fraction from pressure, saturation from content, and content from delivery.
- Test each clause: a correct observation may be followed by the wrong mechanism.
- For "must", "always" or "unchanged", identify the missing condition or give a counterexample.

Tutorial Q# preserves the original question numbering and wording. Answers are reasoned suggestions, not an official key. [Extension] marks material needed for cross-topic questions but not fully covered in these slides. Lecture references use PDF page numbers. Original teaching figures and visible annotations are retained; decorative logos are omitted.

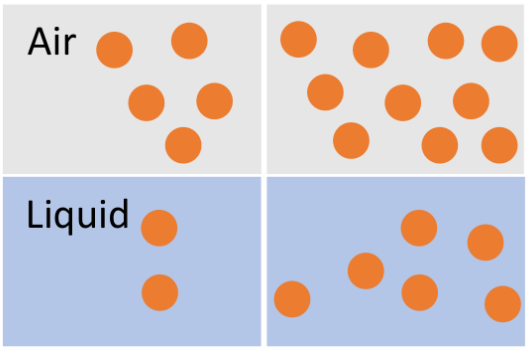
02 / Partial pressure, dissolution and diving

Lecture PDF pp. 4–5, 19 | Tutorial Q13

Law	Relationship	Condition / meaning
Dalton	Pgas = Fgas × Ptotal Ptotal = sum of gas partial pressures	Gas fraction alone does not determine its partial pressure.
Henry	Cdissolved = α × Pgas	At equilibrium, with temperature, gas and solvent fixed.



Original: Dalton's law (lecture p. 4).



Original: Henry's law (lecture p. 5).

- Dry-air PO_2: sea level 760 × 0.21 = 159.6 mmHg; lecture altitude example 250 × 0.21 = 52.5 mmHg. These are not arterial PO_2 values.
- Precision: increasing Pgas raises dissolved concentration; it does not increase α at fixed temperature and medium.

Tutorial Q13 · Original tutorial question

A diver breathing 78% N2 and 21% O2 will have blood dissolved nitrogen level remains unchanged because the fraction of nitrogen is unchanged.

False | The missing variable is total pressure

Depth ↑ → absolute pressure ↑ → PN_2 ↑ despite unchanged FN_2 → equilibrium dissolved N_2 ↑.
[Extension] At about 10 m seawater: 1 → 2 ATA, so dry-gas PN_2 approximately doubles: 0.78 → 1.56 ATA.
Uptake is time-dependent; tissues do not equilibrate immediately.

Reverse the process during ascent

Pressure ↓ → equilibrium dissolved amount ↓ → N_2 leaves solution. Rapid decompression can produce bubbles; slow ascent permits elimination through the lungs.

03 / Altitude and hyperventilation

Lecture PDF pp. 4–5 | Tutorial Q4, Q7 | Alveolar gas and acid–base equations are extensions

Altitude chain

PB ↓ at unchanged FiO_2 → PIO_2 ↓ → PAO_2 ↓ → reduced O_2 gradient into blood.

[Extension] Relationship	Interpretation
$PIO_2 = FiO_2 \times (PB - 47)$	Humidified inspired gas at 37°C; pressures in mmHg.
$PAO_2 \approx PIO_2 - PaCO_2/R$; $R \approx 0.8$	Simplified alveolar gas equation under usual room-air conditions.
Alveolar ventilation ↑ at fixed CO_2 production	$PaCO_2$ ↓ → PAO_2 ↑, without changing FiO_2.

Tutorial Q4 · Original tutorial question

Hyperventilation at high altitude helps improve oxygenation even though it causes respiratory acidosis.

False | Correct benefit, wrong acid–base label

Hyperventilation can improve oxygenation, but acutely $PaCO_2$ ↓ → H^+ ↓ → pH ↑: respiratory alkalosis.
Later renal HCO_3^- loss is compensation; it does not change the name of the primary respiratory disturbance.

Tutorial Q7 · Original tutorial question

Hyperventilation can increase arterial PO_2 even when oxygen concentration in inspired air remains unchanged.

True under usual physiological conditions

FiO_2 unchanged → alveolar ventilation ↑ → $PaCO_2$ ↓ → PAO_2 ↑ → PaO_2 can rise.
PAO_2 = alveolar; PaO_2 = arterial. Shunt or severe gas-exchange impairment can limit the response.

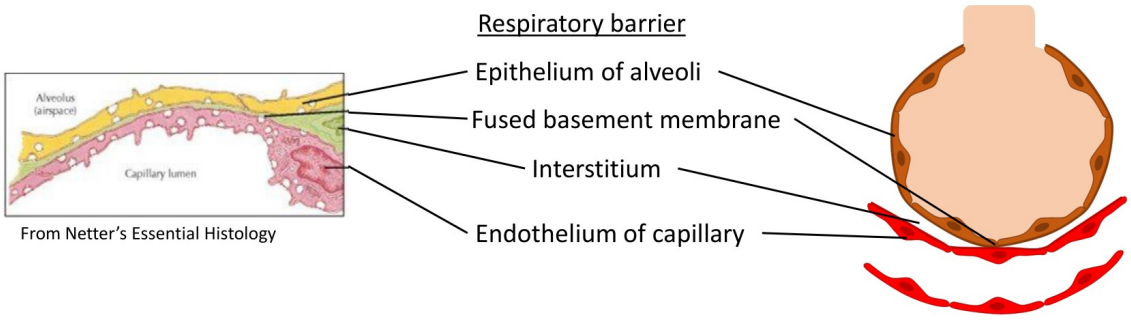
Key comparison: hyperventilation may improve PaO_2 relative to less ventilation at the same altitude; it does not guarantee restoration of the sea-level value.

04 / Diffusion: apply Fick's law

Lecture PDF pp. 7–9, 19

Equation and barrier

$\dot{V}gas = A \times D \times \Delta P / T$; $D \propto$ solubility coefficient / $\sqrt{}$molecular weight.
Gas must dissolve before crossing the barrier, and requires a partial-pressure gradient.
Alveolar epithelium → fused basement membrane / thin interstitium → capillary endothelium.



What determines the rate of diffusion of the gases across the membrane?

Original respiratory barrier (lecture p. 7; Netter's Essential Histology).

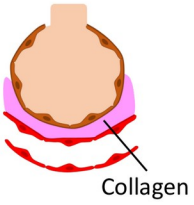
Factor	Effect / application
A: effective surface area	Alveolar wall destruction in emphysema → A ↓ → diffusion ↓.
T: diffusion distance	T ↑ in fibrosis or edema → diffusion ↓.
ΔP: pressure gradient	ΔP ↑ → diffusion ↑; compare the same gas across the barrier.
D: diffusion coefficient	CO_2 diffuses about 20× more readily than O_2, mainly because it is more soluble.

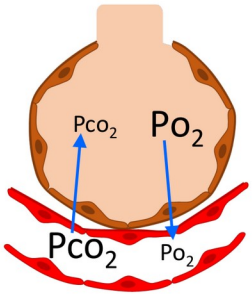
$$V = \frac{A \times D \times \Delta P}{T}$$

V = rate of gas diffusion
A = surface area for diffusion
D = Diffusivity
ΔP = pressure gradient
T = thickness of membrane

CO_2 20x O_2

$$D \propto \frac{solubility}{\sqrt{MW}}$$





Original Fick diagram and pressure-gradient directions (lecture p. 8).

- Normal comparison: for a given gas and healthy barrier, D and T change relatively little; changes in A and ΔP therefore explain much of the change in transfer rate.

$\dot{V}$ is diffusion rate, not minute ventilation. The molecular-weight term alone does not predict diffusivity: CO_2 is heavier than O_2 but much more soluble.

05 / Diffusion applications and slide exercise

Lecture PDF pp. 9, 19 | Original exercise with typed answers below

$$V = \frac{A \times D \times \Delta P}{T}$$

V = rate of gas diffusion
A = surface area for diffusion
D = Diffusivity
ΔP = pressure gradient
T = thickness of membrane

Choose the words to fill in the blank:

Surface area Oxygen Thickness of membrane

Diffusivity Pressure gradient Carbon dioxide

Normally, thickness & diffusivity don't change significantly, therefore the rate of diffusion (V) depends on _____ and _____. In general, _____ is more soluble so it can be eliminated faster, while partial pressure of _____ is high so it can be taken up quickly.

Pathologically, _____ is reduced in emphysema, _____ is increased in fibrosis.

Original in-slide fill-in exercise, retaining the existing annotations (lecture p. 9).

Original blank / comparison	Answer and interpretation
Normally do not change significantly	Thickness; diffusivity. Fixed for the simplified comparison, not universal constants.
Main normal determinants	Surface area; pressure gradient.
More soluble; eliminated readily	Carbon dioxide: high solubility supports rapid exchange despite a smaller usual pressure gradient.
Taken up readily	Oxygen: a favourable alveolar-to-blood PO_2 gradient drives uptake.
Emphysema / fibrosis	Surface area decreases / membrane thickness increases; both lower diffusion capacity.

Use ratios rather than verbal guesses

$\dot{V}new/\dot{V}old = (Anew/Aold) \times (Dnew/Dold) \times (\Delta Pnew/\Delta Pold) \times (Told/Tnew)$.

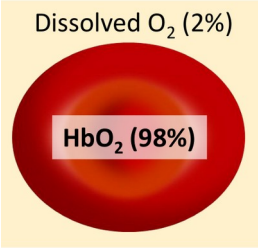
If A halves and T doubles, with other factors fixed: $\dot{V}$ falls to 1/4.

If ΔP then doubles as well, the final rate is 1/2 of baseline: combine all factors before judging.

06 / O_2: from saturation to delivery

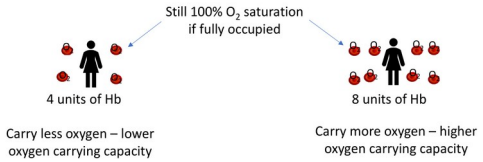
Lecture PDF pp. 11–12 | Tutorial Q3 | Content and delivery equations are extensions

Variable	What it measures
PaO_2	Partial pressure of dissolved arterial O_2; not total oxygen content.
SaO_2 / SpO_2	Fraction of Hb O_2 sites occupied / pulse-oximeter estimate; not Hb quantity.
O_2 capacity	Maximum Hb-bound O_2 at full saturation; set by Hb concentration, not current saturation.
CaO_2 / DO_2	O_2 per dL arterial blood / O_2 delivered per minute.



Oxygen content

Original O_2 forms (p. 11).



Original: Hb amount differs; saturation can stay 100% (p. 12).

- About 98% of blood O_2 is Hb-bound (oxyhemoglobin), ~2% dissolved; the proportions vary with conditions. Dissolved O_2 sets PO_2, which determines saturation along the dissociation curve.

Tutorial Q3 · Original tutorial question

A person with normal oxygen saturation must have normal oxygen delivery to tissues.

False | Saturation does not measure Hb quantity or blood flow
Hb ↓ → CaO_2 ↓; cardiac output ↓ → DO_2 ↓ even with normal CaO_2.
$CaO_2 \approx 1.34 \times Hb \times SaO_2 + 0.003 \times PaO_2$; $DO_2 = CO \times CaO_2 \times 10$.
Units: Hb g/dL; SaO_2 as a fraction; PaO_2 mmHg; CaO_2 mL/dL; CO L/min; DO_2 mL/min.

Integrated practice 1 | Self-written; not an original exam question
Hb falls from 15 to 8 g/dL. SaO_2 stays 0.98 and PaO_2 stays 100 mmHg; CO rises from 5 to 7 L/min. Has delivery been fully restored?
Reasoning: No. CaO_2: 20.0 → 10.8 mL/dL; DO_2: about 1,000 → 756 mL/min. The flow increase only partly compensates. CO would need to be about 9.3 L/min to restore baseline delivery.

- Each Hb has four heme groups containing Fe^{2+}; each binds one O_2. Capacity ≈ 1.34 × Hb; actual content also depends on saturation.

07 / Anemia, cyanosis and hypoxia

Lecture PDF p. 13 | Tutorial Q6 | Hypoxia and compensation details are extensions

Term	Distinction
Anemia	Low Hb concentration → reduced carrying capacity. Common causes in the slide: iron or vitamin B_{12} deficiency, and hemolysis.
Cyanosis	Blue/purple discoloration of lips, tongue or extremities. Lecture threshold: absolute deoxyHb approximately >5 g/dL, not a fixed saturation percentage.
Hypoxemia	Reduced arterial PaO_2.
Tissue hypoxia	Inadequate tissue O_2 supply or use; depends on content, perfusion and metabolic demand.



Original skin-colour illustration (lecture p. 13).

In-slide questions

- Anemia must cause cyanosis: False. Low total Hb may prevent enough absolute deoxyHb accumulating for visible cyanosis.
- Cyanosis must imply anemia: False. Poor oxygenation or slow local flow can cause cyanosis with normal Hb.

Tutorial Q6 · Original tutorial question

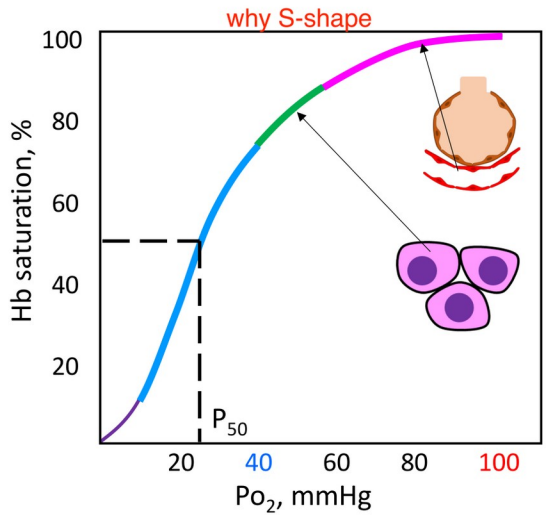
A person with hypoxemia will have hypoxia; But a person with hypoxia may not have hypoxemia.

Separate the two directions

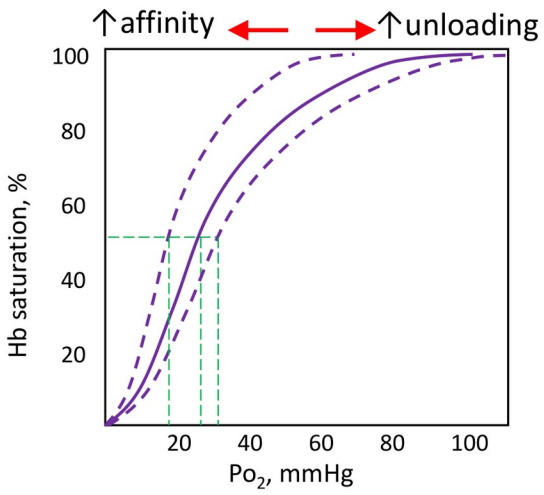
• Hypoxia without hypoxemia: possible in anemia, low perfusion or impaired cellular O_2 use.
• Hypoxemia inevitably causing hypoxia: too absolute. Increased flow or extraction may compensate for mild hypoxemia.
Strict literal reading: the whole statement can be False. Introductory teaching may use True as a simplification; state the compensation caveat. No official answer key was supplied.

08 / Dissociation curve and P_{50}

Lecture PDF pp. 14–15



Original sigmoid O_2 curve and P_{50} (p. 14).



Original affinity / unloading shifts (p. 15).

Feature	Interpretation
Sigmoid shape	Positive cooperativity: binding one O_2 facilitates subsequent binding. Releasing O_2 also facilitates further unloading.
Pulmonary plateau: PO_2 ~60–100	Moderate PO_2 falls cause relatively small saturation changes. Typical arterial point: PaO_2 ~100 mmHg, SaO_2 ~98%.
Middle region: PO_2 ~40–60	Steeper than the plateau; decreasing PO_2 promotes O_2 unloading.
Steep lower region: PO_2 ~15–40	A small PO_2 fall can release substantial O_2. Typical resting mixed venous point: PvO_2 ~40 mmHg, SvO_2 ~75%.
P_{50} normally ~26.5 mmHg	PO_2 needed for 50% saturation; higher P_{50} means lower affinity.
Right / left shift	Right: P_{50} ↑, affinity ↓, easier unloading. Left: the reverse. Compare at the same PO_2.

Answers to the original P_{50} fill-in prompts

P_{50} increased → affinity decreased → easier unloading.
P_{50} decreased → affinity increased → harder unloading.
P_{50} is a pressure at 50% saturation, not a percentage of the normal PO_2.

09 / Curve shifts: factors and mechanisms

Lecture PDF p. 16 | Original annotated table preserved with corrections below

H+ cause comformational change of haemoglobin

Factors	Shift to	Explanation
↓pH/ ↑Pco_2	Right	High CO_2 indicates needs for O_2; CO_2 produces H^+, so: ↓pH → ↓affinity to unload O_2 (Bohr effect)
↑2,3-BPG#	Right	High 2,3-BPG indicates active metabolism, 2,3-BPG is acidic, so: ↓pH → ↓affinity to unload O_2 (Bohr effect) fetus hae is insensitive to 23BPG
↑temperature	Right	High temperature indicates active metabolism, so ↓affinity to unload O_2

#2,3-BPG = 2,3-bisphophoglycerate, products of glycolysis in erythrocytes

Original right-shift factors and annotations (lecture p. 16); mechanism clarification below.

Change	Shift / mechanism	Physiological meaning
PCO_2 ↑ / pH ↓	Right; H^+ changes Hb conformation and lowers O_2 affinity (Bohr).	Working tissues generate CO_2/H^+, favouring unloading.
Temperature ↑	Right; lower O_2 affinity.	Warm, metabolically active tissue can receive more O_2.
2,3-BPG ↑	Right; binds deoxyHb and stabilises the T state.	2,3-bisphosphoglycerate is a product of RBC glycolysis.
Opposite changes	Generally shift left: affinity ↑, P_{50} ↓.	Facilitates binding at the same PO_2 but makes unloading harder.

Keep the directions; distinguish the mechanisms

- 2,3-BPG stabilises deoxyHb in the T state → right shift; it is not simply “acid causes Bohr effect”.
- [Extension] HbF binds 2,3-BPG less strongly → higher O_2 affinity; not complete insensitivity.

Use exercise to reconstruct the logic

Metabolic demand ↑ → CO_2/H^+ and temperature ↑ → Hb affinity ↓ → right shift
→ more unloading at the same tissue PO_2. A shift describes affinity, not a change in Hb concentration.

10 / Curve applications: lung loading and CO

Lecture PDF pp. 14–16 | Tutorial Q5, Q10 | CO-specific mechanisms are extensions

Tutorial Q5 · Original tutorial question

A right shift of the oxygen dissociation curve reduces oxygen uptake by the lungs.

Condition-dependent | Specify pulmonary PO_2

At fixed PO_2, a right shift lowers saturation and can reduce loading.
On the normal pulmonary plateau, a moderate shift usually has a small loading penalty; enhanced tissue unloading is more prominent. At lower alveolar PO_2, the penalty matters more.
Do not claim "right shifts never reduce loading". An unconditional T/F answer needs an assumed PO_2 and shift magnitude; no official key was supplied.

Tutorial Q10 · Original tutorial question

Carbon monoxide poisoning competes with O2 to bind to Hb. Therefore, it decreases Hb saturation and Hb affinity for O2.

False | The affinity direction is reversed

CO occupies heme sites → fewer sites available for O_2 → effective capacity and CaO_2 ↓.
Remaining available sites bind O_2 more strongly → left shift → impaired tissue unloading.
The combined problem is reduced carriage plus impaired release, not reduced affinity.

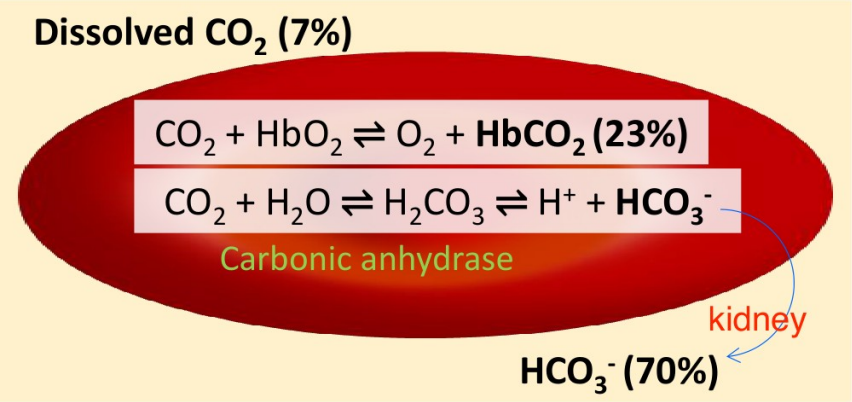
Measurement	Why it can mislead in CO poisoning
PaO_2	May be normal: it measures dissolved O_2, not CO occupancy of Hb.
Conventional SpO_2	Can appear normal/high; standard two-wavelength instruments cannot reliably distinguish COHb.
O_2 saturation	Functional and fractional saturation differ with COHb; a reassuring number does not guarantee normal CaO_2.

Connection to Q3: neither normal PaO_2 nor a normal-looking pulse-oximeter reading establishes adequate O_2 content or delivery.

11 / CO_2 forms and carbonic anhydrase

Lecture PDF p. 17 | Tutorial Q12 partly overlaps; renal drug mechanism is an extension

Form	Approximate share	Key point
Dissolved CO_2	7%	Determines blood PCO_2.
Carbamino-Hb: $HbCO_2$	23%	CO_2 binds globin amino groups, not heme iron.
Bicarbonate: HCO_3^-	70%	Formed rapidly in RBCs; much is transported in plasma.



Original CO_2 transport forms, including the "kidney" annotation (lecture p. 17).

Reversible reaction

$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$.

Carbonic anhydrase accelerates reversible hydration/dehydration; it does not change the equilibrium constant.

- Carbamino-Hb = CO_2 bound to globin. Carboxy-Hb = CO bound to heme; the latter impairs O_2 transport.

Tutorial Q12 · Original tutorial question

Carbonic anhydrase inhibitor stimulates hyperventilation by promoting bicarbonate ion excretion.

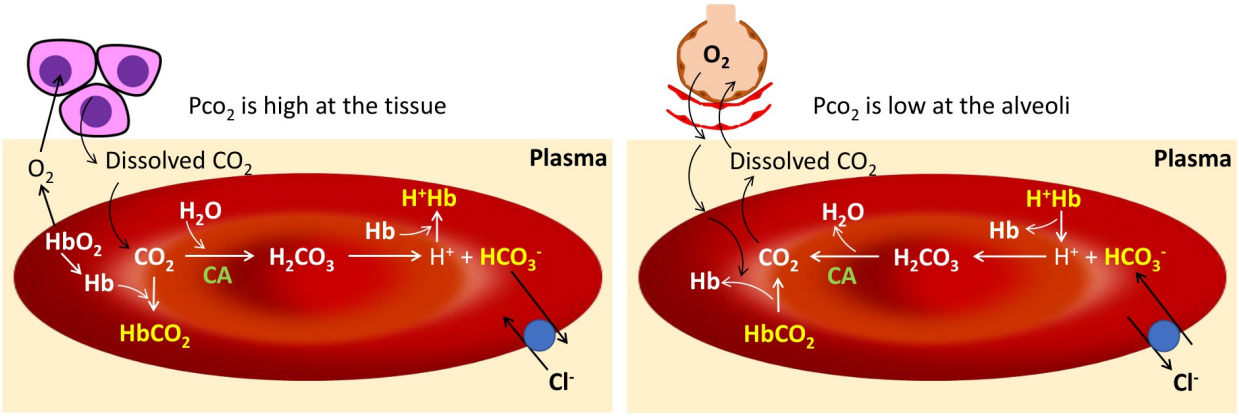
True | Primarily a renal mechanism [Extension]

Acetazolamide → renal tubular CA inhibition → HCO_3^- reabsorption ↓ → urinary HCO_3^- loss ↑ → metabolic acidosis → ventilatory drive ↑. At altitude, this can reduce the inhibitory effect of respiratory alkalosis.

The explanation is renal bicarbonate loss, not increased RBC CO_2 production.

12 / CO_2 exchange: Bohr versus Haldane

Lecture PDF pp. 16–18



18

Original tissue end (left) and pulmonary end (right), lecture p. 18.

Process	Tissue capillaries	Pulmonary capillaries
CO_2 gradient	High tissue PCO_2 → CO_2 enters blood/RBC.	Lower alveolar PCO_2 → CO_2 leaves blood.
CA reaction	$CO_2 + H_2O \rightarrow H^+ + HCO_3^-$.	$H^+ + HCO_3^- \rightarrow CO_2 + H_2O$.
Hb buffering	DeoxyHb binds H^+ more readily.	Hb oxygenation promotes H^+ release.
Chloride shift	HCO_3^- out of RBC; Cl^- in.	HCO_3^- into RBC; Cl^- out. The exchange preserves electroneutrality.
Carbamino transport	DeoxyHb favours CO_2 binding.	Oxygenation favours CO_2 release.

Name the effect by its input → output

Bohr: CO_2 / H^+ ↑ → O_2 affinity ↓ → tissue O_2 unloading.

Haldane: Hb deoxygenation → greater CO_2/H^+ binding; oxygenation → CO_2 unloading.

They cooperate at tissues and lungs, but describe different causal directions.

Integrated practice 2 | Self-written; not an original exam question

At the same PCO_2, oxygenated blood carries less total CO_2 than deoxygenated blood. Explain why this does not contradict Henry's law, and predict the pulmonary chloride shift.

Reasoning: Henry fixes only dissolved CO_2 when α and PCO_2 are fixed. Oxygenation reduces carbamino binding and releases H^+, driving HCO_3^- towards CO_2; total CO_2 content can therefore fall (Haldane). In lungs, HCO_3^- enters RBCs while Cl^- leaves. A lower alveolar PCO_2 supports net elimination; oxygenation is not the pressure gradient itself.